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Lab Experiments.

Set 1

Program 1: Line Chart for Monthly Sales

```
month <- c("January", "February", "March", "April", "May")
```

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

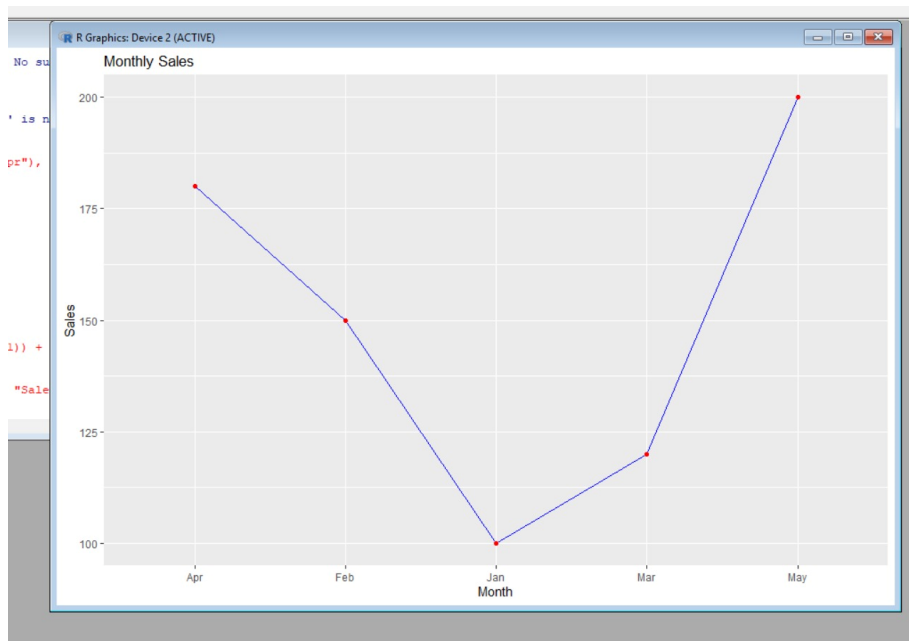
```
plot(sales,  
      type = "o",  
      xaxt = "n",  
      xlab = "Month",  
      ylab = "Sales ($)",  
      main = "Monthly Sales",  
      col = "blue",  
      lwd = 2,  
      pch = 16)
```

```
axis(1, at = 1:5, labels = month)
```

```
grid()
```

Output:

Output:



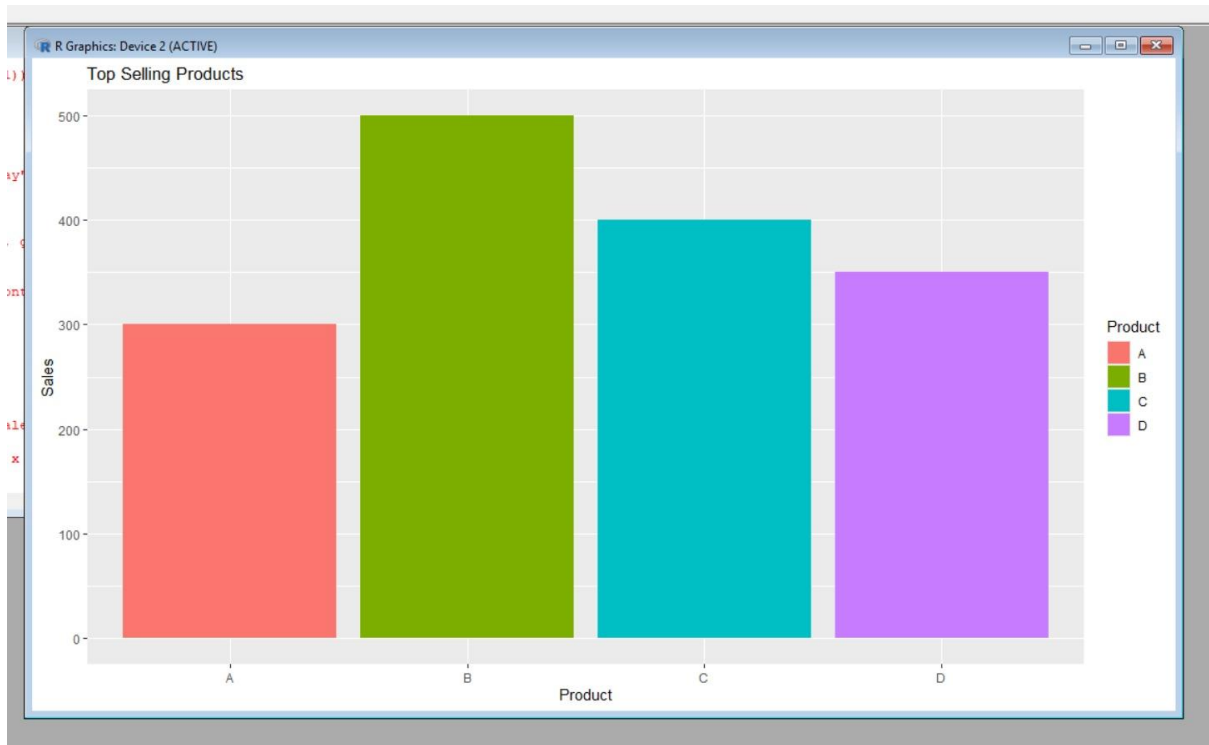
Program 2: Bar Chart for Top-Selling Products

```
products <- c("Laptop", "Mobile", "Tablet", "Headphones", "Smartwatch")
```

```
sales <- c(120, 180, 90, 150, 100)
```

```
barplot(sales,  
        names.arg = products,  
        col = "skyblue",  
        main = "Top Selling Products",  
        xlab = "Products",  
        ylab = "Units Sold",  
        border = "black")
```

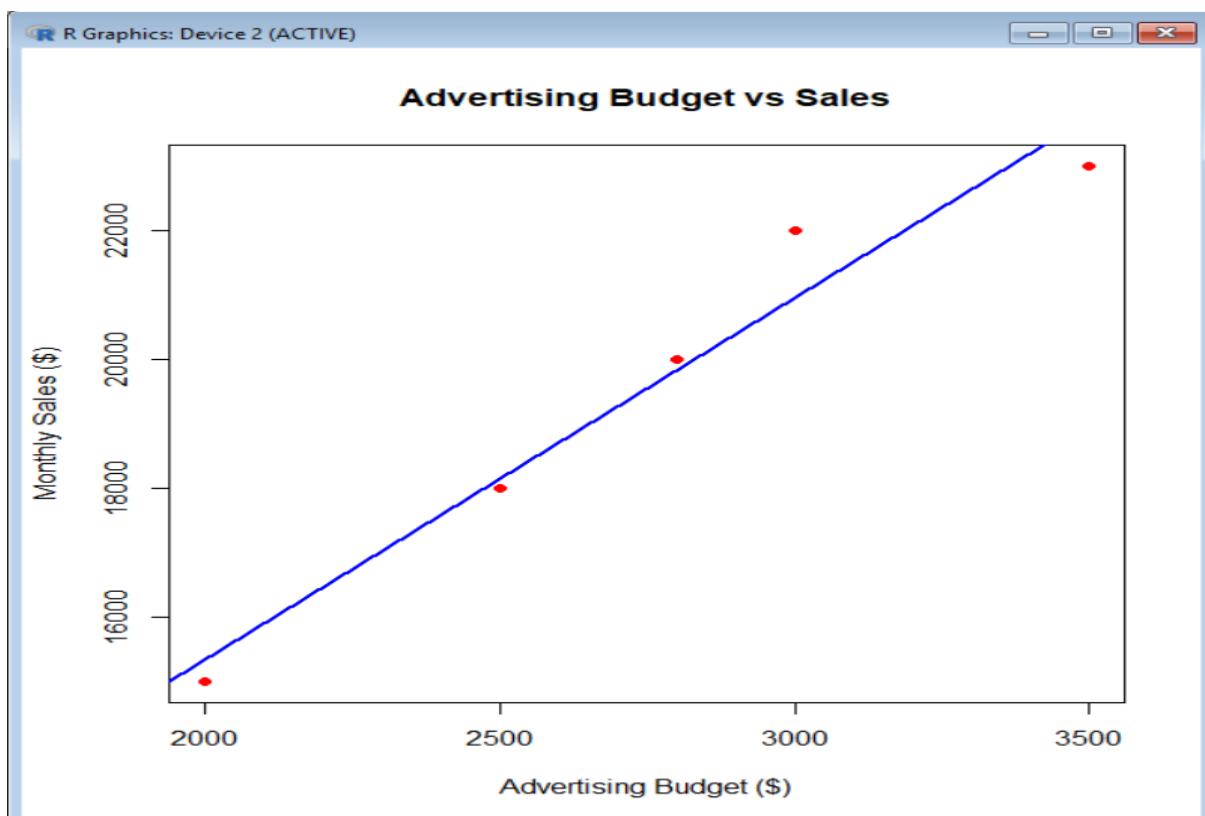
Output:



Program 3: Scatter Plot (Advertising Budget vs Monthly Sales)

```
advertising <- c(2000, 2500, 3000, 2800, 3500)
sales <- c(15000, 18000, 22000, 20000, 23000)
plot(advertising,
     sales,
     main = "Advertising Budget vs Sales",
     xlab = "Advertising Budget ($)",
     ylab = "Monthly Sales ($)",
     pch = 19,
     col = "red")
abline(lm(sales ~ advertising), col = "blue", lwd = 2)
```

Output:



Program 4: Interactive Dashboard (Line Chart + Bar Chart)

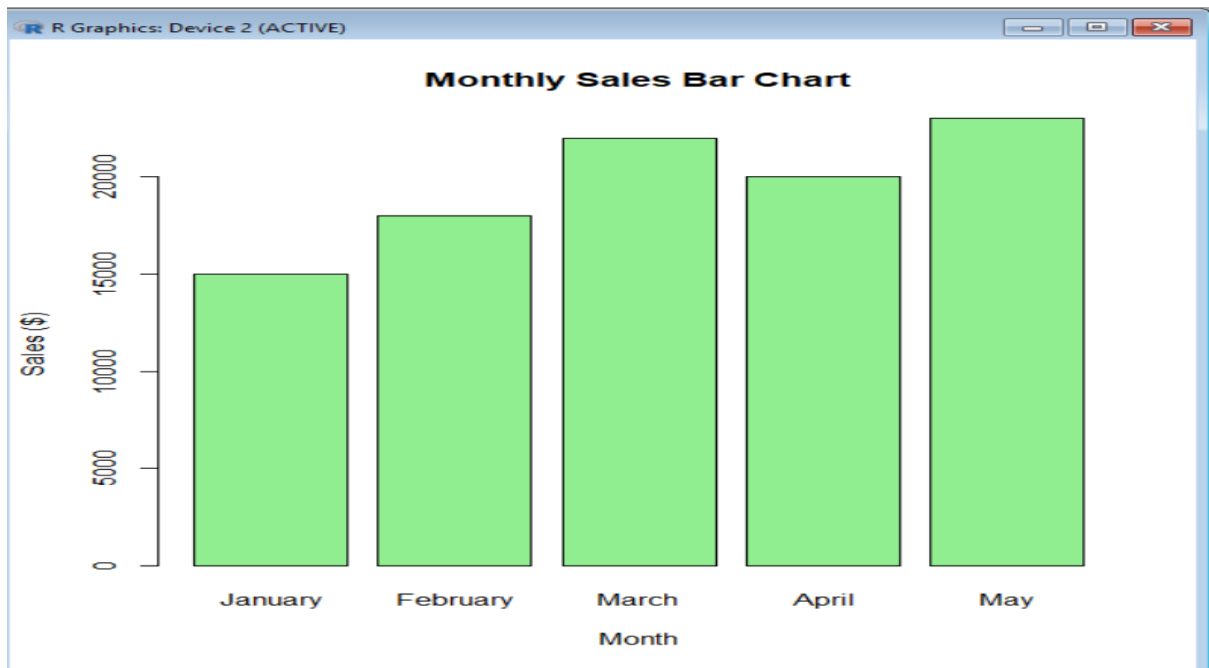
```
install.packages("manipulate")

library(manipulate)

month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

manipulate(
{
  par(mfrow = c(1,2))
  plot(sales,
        type = "o",
        xaxt = "n",
        xlab = "Month",
        ylab = "Sales ($)",
        main = "Monthly Sales",
        col = "blue",
        lwd = 2,
        pch = 16)
  axis(1, at = 1:5, labels = month)
  barplot(sales,
           names.arg = month,
           col = "lightgreen",
           main = "Monthly Sales Bar Chart",
           xlab = "Month",
           ylab = "Sales ($)")
},
dummy = slider(1,5,initial = 1))
```

Output:



Set 2

Program 1: Histogram of Customer Ages

```
age <- c(25, 30, 35, 28, 40)
```

```
hist(age,
```

```
  main = "Distribution of Customer Ages",
```

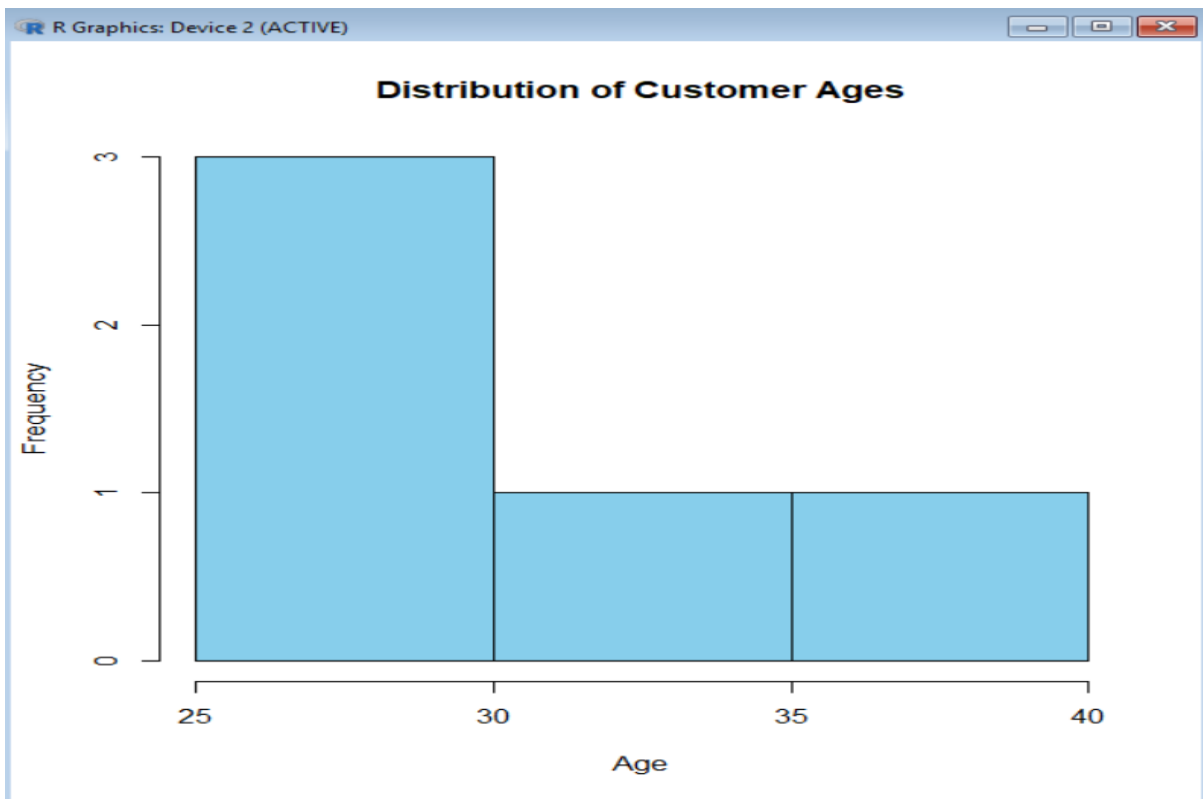
```
  xlab = "Age",
```

```
  ylab = "Frequency",
```

```
  col = "skyblue",
```

```
  border = "black")
```

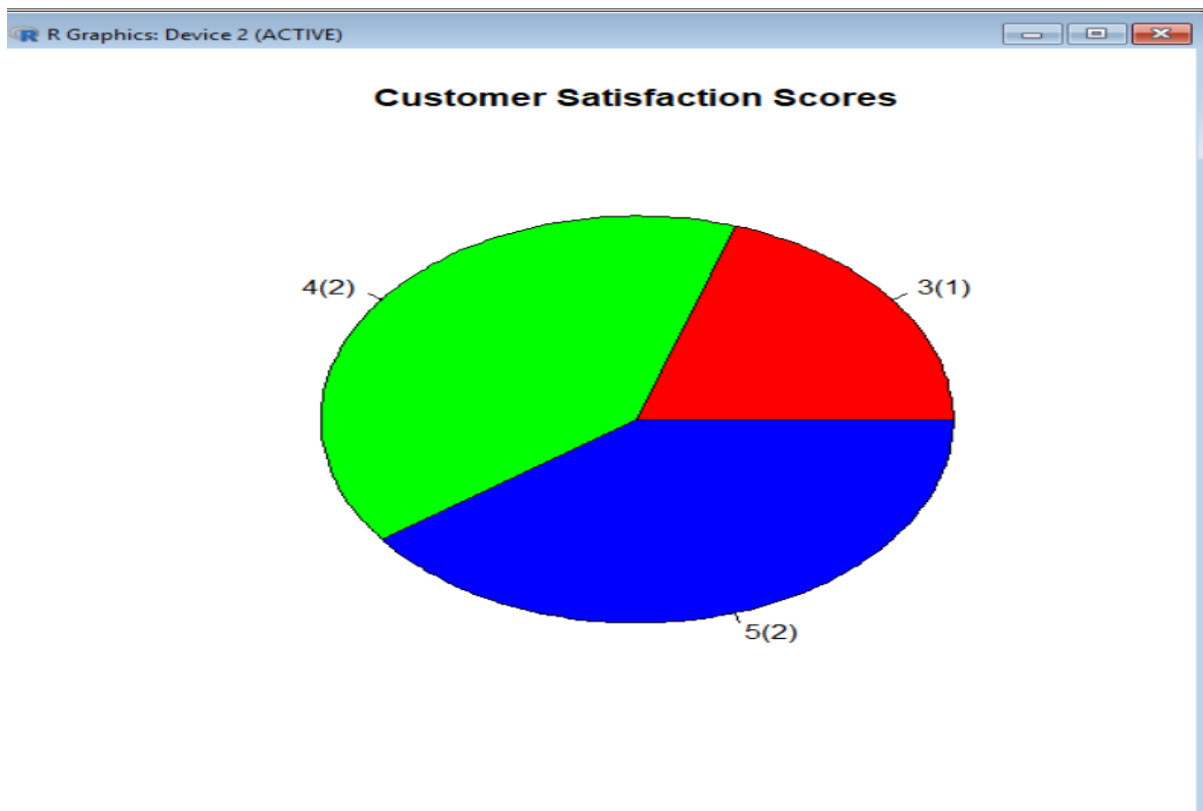
Output:



Program 2: Pie Chart of Customer Satisfaction Scores

```
customer_id <- c(1,2,3,4,5)
age <- c(25,30,35,28,40)
satisfaction <- c(4,5,3,4,5)
data <- data.frame(customer_id, age, satisfaction)
score_count <- table(data$satisfaction)
pie(score_count,
     main="Customer Satisfaction Scores",
     labels=paste(names(score_count), "(", score_count, ")", sep=""),
     col=rainbow(length(score_count)))
```

Output:



3. Stacked Bar Chart of Satisfaction Scores by Age Group

```
customer_id <- c(1,2,3,4,5)
age <- c(25,30,35,28,40)
satisfaction <- c(4,5,3,4,5)
data <- data.frame(customer_id, age, satisfaction)
data$AgeGroup <- cut(data$age,
  breaks=c(20,30,40,50),
  labels=c("21-30","31-40","41-50"),
  include.lowest=TRUE)
table_data <- table(data$AgeGroup, data$satisfaction)
```

```
barplot(table_data,  
        main="Satisfaction Scores by Age Group",  
        xlab="Age Group",  
        ylab="Number of Customers",  
        col=rainbow(ncol(table_data)),  
        legend=colnames(table_data))
```

Output:

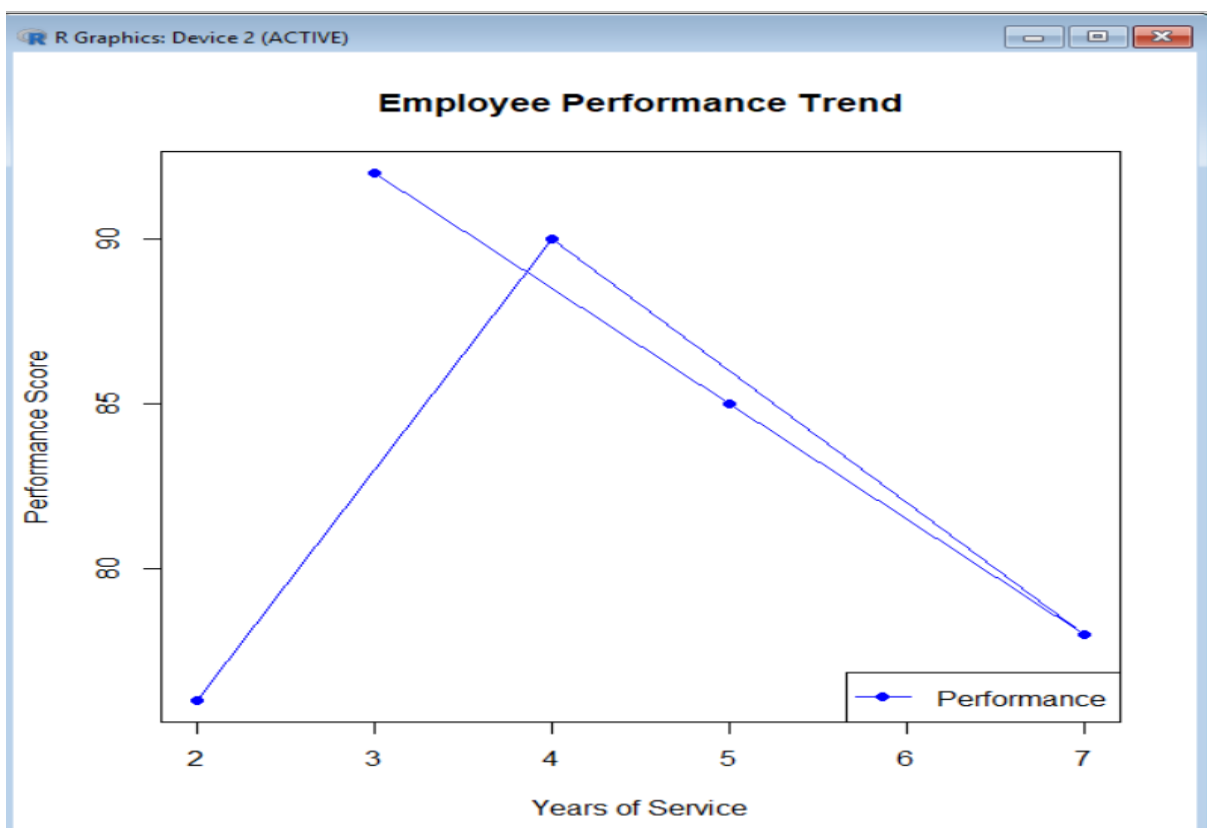


Set 3

Program 1: Line Chart – Performance Trend of Employees


```
EmployeeID <- c(1,2,3,4,5)
Department <- c("Sales","HR","Marketing","Sales","HR")
YearsService <- c(5,3,7,4,2)
PerformanceScore <- c(85,92,78,90,76)
data <- data.frame(EmployeeID, Department, YearsService, PerformanceScore)
plot(data$YearsService,
     data$PerformanceScore,
     type="o",
     col="blue",
     pch=16,
     xlab="Years of Service",
     ylab="Performance Score",
     main="Employee Performance Trend")
legend("bottomright",
     legend="Performance",
     col="blue",
     lty=1,
     pch=16
```

Output:

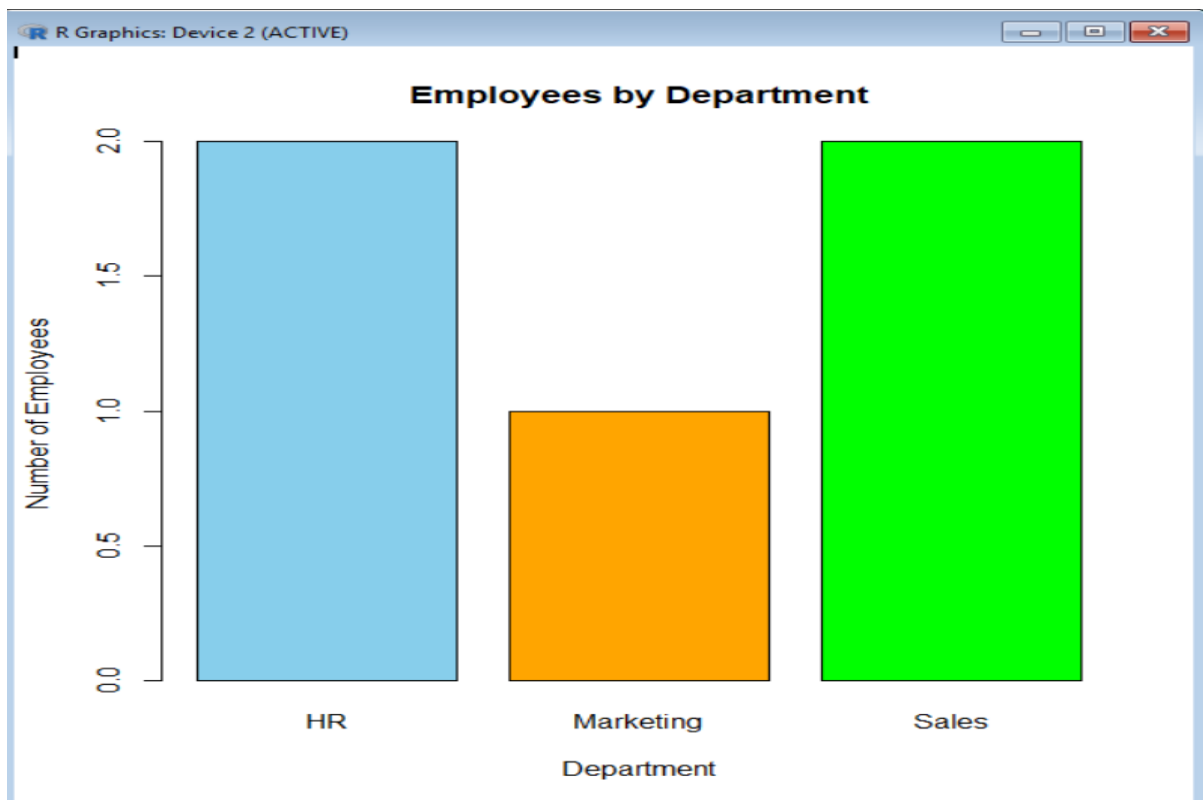


Program 2: Bar Chart – Employees by Department

```
EmployeeID <- c(1,2,3,4,5)
Department <- c("Sales","HR","Marketing","Sales","HR")
YearsService <- c(5,3,7,4,2)
PerformanceScore <- c(85,92,78,90,76)
data <- data.frame(EmployeeID, Department, YearsService, PerformanceScore)
dept_count <- table(data$Department)

barplot(dept_count,
        col=c("skyblue","orange","green"),
        main="Employees by Department",
        xlab="Department",
        ylab="Number of Employees")
```

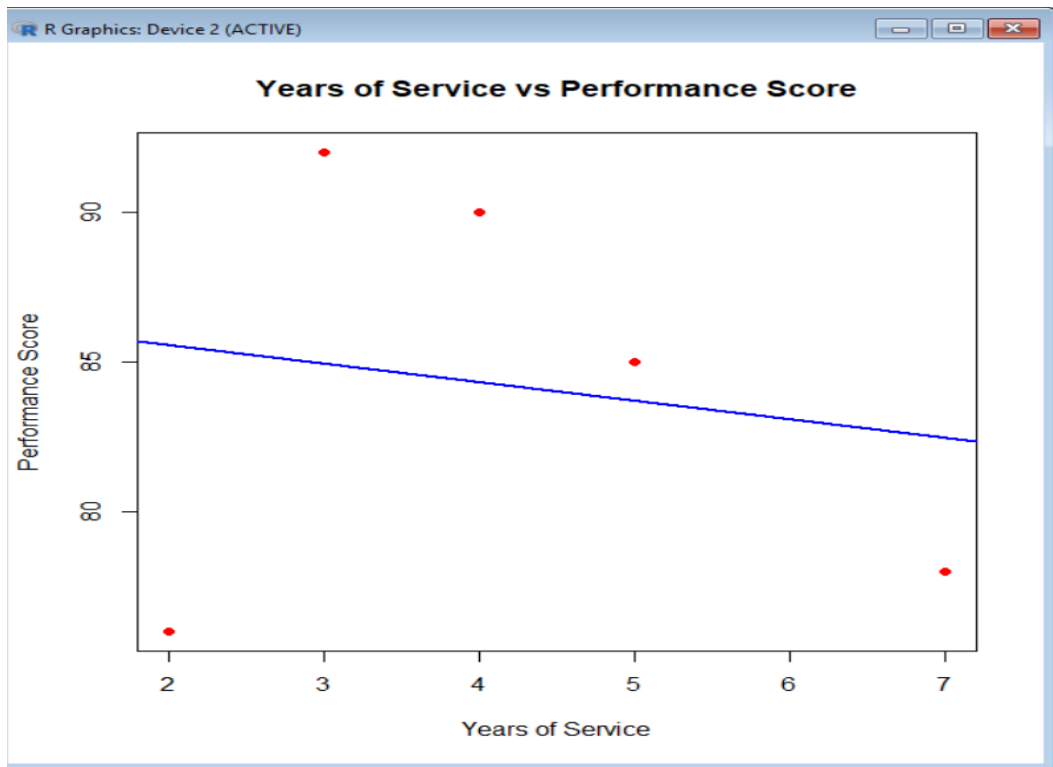
Output:



Program 3: Scatter Plot – Years of Service vs Performance Score

```
EmployeeID <- c(1,2,3,4,5)
Department <- c("Sales","HR","Marketing","Sales","HR")
YearsService <- c(5,3,7,4,2)
PerformanceScore <- c(85,92,78,90,76)
data <- data.frame(EmployeeID, Department, YearsService, PerformanceScore)
plot(data$YearsService,
      data$PerformanceScore,
      pch=19,
      col="red",
      xlab="Years of Service",
      ylab="Performance Score",
      main="Years of Service vs Performance Score")
abline(lm(PerformanceScore ~ YearsService, data=data),
      col="blue",
      lwd=2)
```

Output:



Program 4: Interactive Dashboard (Using plotly)

```
library(plotly)
```

```
EmployeeID <- c(1,2,3,4,5)
```

```
Department <- c("Sales","HR","Marketing","Sales","HR")
```

```
YearsService <- c(5,3,7,4,2)
```

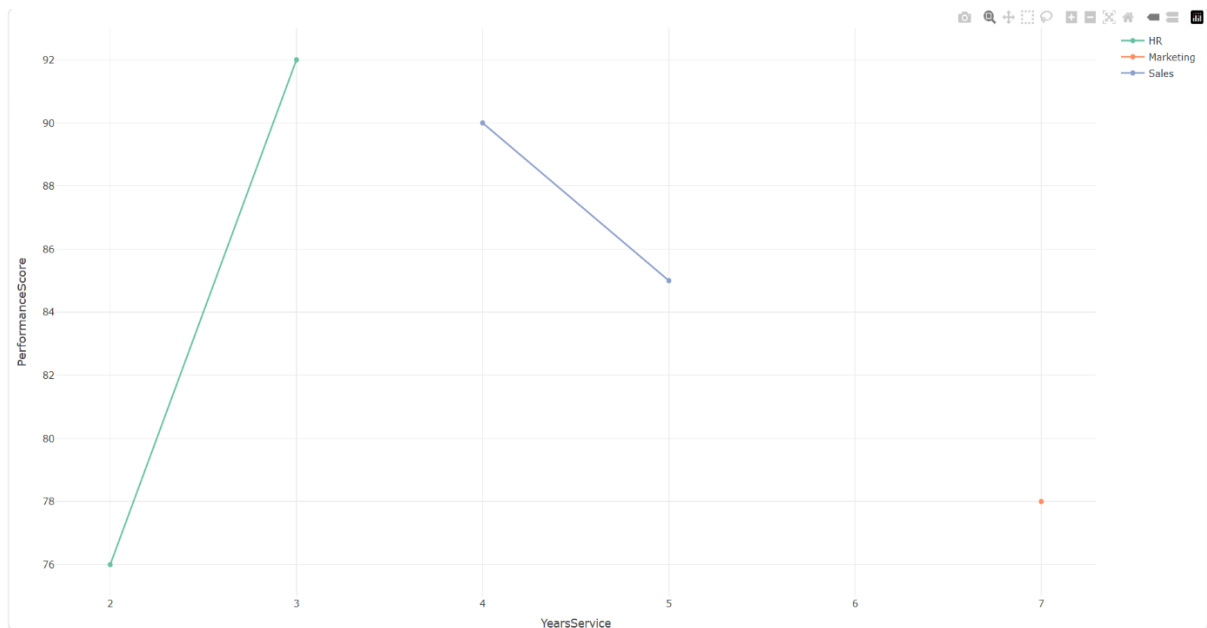
```
PerformanceScore <- c(85,92,78,90,76)
```

```
data <- data.frame(EmployeeID, Department, YearsService, PerformanceScore)
```

```
fig <- plot_ly(data,  
  x=~YearsService,  
  y=~PerformanceScore,  
  type="scatter",  
  mode="lines+markers",  
  color=~Department)
```

```
fig
```

Output:



Set 4

Program 1: Bar Chart – Quantity of Each Product

```
ProductID <- c(1,2,3,4,5)
```

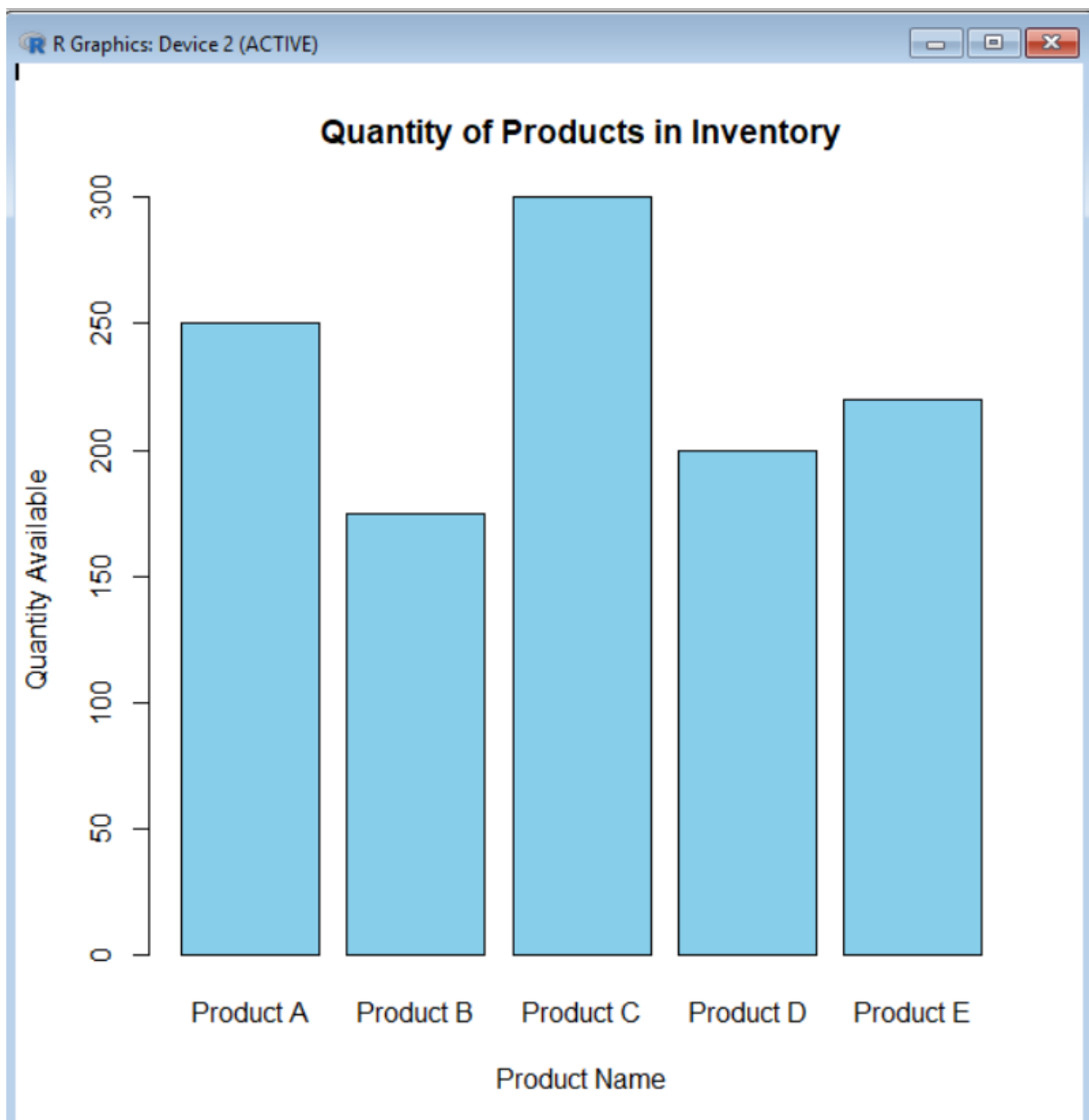
```
ProductName <- c("Product A","Product B","Product C","Product D","Product E")
```

```
QuantityAvailable <- c(250,175,300,200,220)
```

```
data <- data.frame(ProductID, ProductName, QuantityAvailable)
```

```
barplot(data$QuantityAvailable,  
        names.arg=data$ProductName,  
        col="skyblue",  
        main="Quantity of Products in Inventory",  
        xlab="Product Name",  
        ylab="Quantity Available")
```

Output:



Program 2: Stacked Bar Chart – Product Quantity by Category

```
ProductID <- c(1,2,3,4,5)
```

```
ProductName <- c("Product A","Product B","Product C","Product D","Product E")
```

```
QuantityAvailable <- c(250,175,300,200,220)
```

```
Category <- c("Electronics","Electronics","Furniture","Furniture","Electronics")
```

```
data <- data.frame(ProductID, ProductName, QuantityAvailable, Category)
```

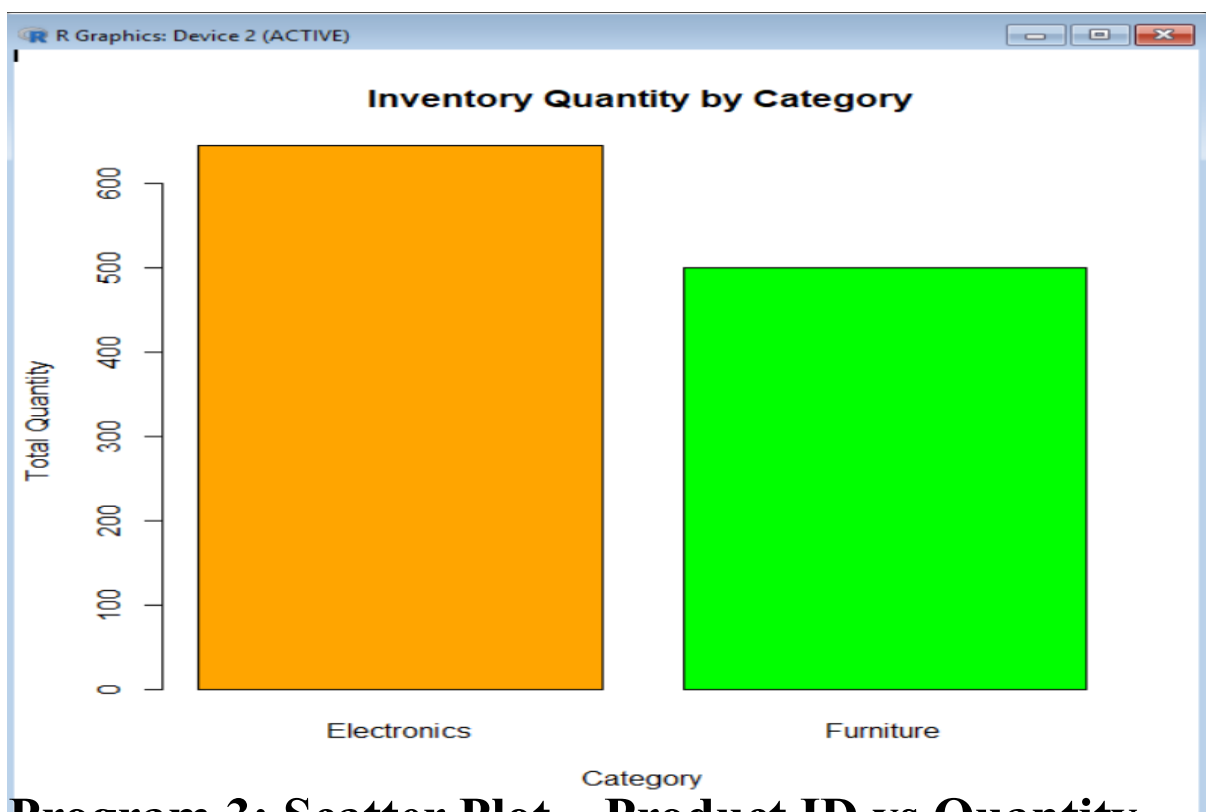
```
table_data <- tapply(data$QuantityAvailable,
```

```

        data$Category,
        sum)
barplot(table_data,
        col=c("orange","green"),
        main="Inventory Quantity by Category",
        xlab="Category",
        ylab="Total Quantity")

```

Output:



Program 3: Scatter Plot – Product ID vs Quantity Available

```

ProductID <- c(1,2,3,4,5)
ProductName <- c("Product A","Product B","Product C","Product D","Product E")
QuantityAvailable <- c(250,175,300,200,220)
data <- data.frame(ProductID, ProductName, QuantityAvailable)
plot(data$ProductID,
      data$QuantityAvailable,
      pch=19,

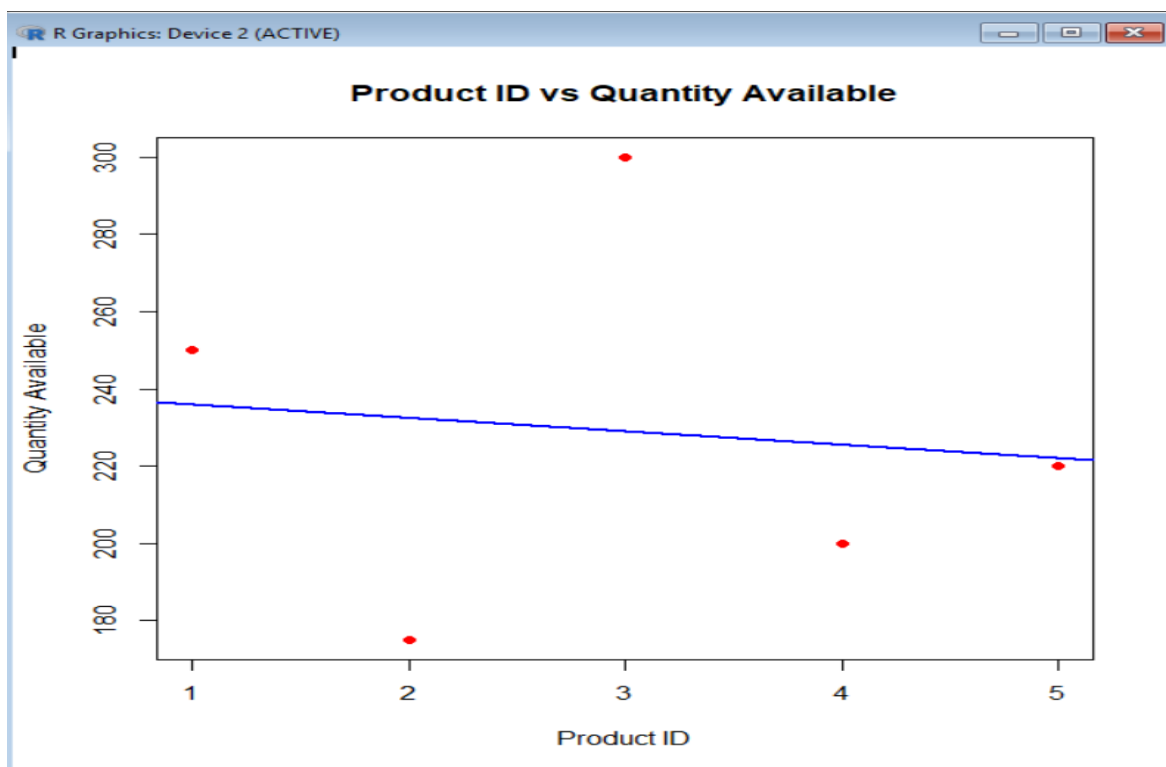
```

```

col="red",
xlab="Product ID",
ylab="Quantity Available",
main="Product ID vs Quantity Available")
abline(lm(QuantityAvailable ~ ProductID, data=data),
      col="blue",
      lwd=2)

```

Output:



Program 4: Interactive Dashboard (Plotly)

```

library(plotly)
ProductID <- c(1,2,3,4,5)
ProductName <- c("Product A","Product B","Product C","Product D","Product E")
QuantityAvailable <- c(250,175,300,200,220)
data <- data.frame(ProductID, ProductName, QuantityAvailable)
fig <- plot_ly(data,
               x=~ProductName,
               y=~QuantityAvailable,

```



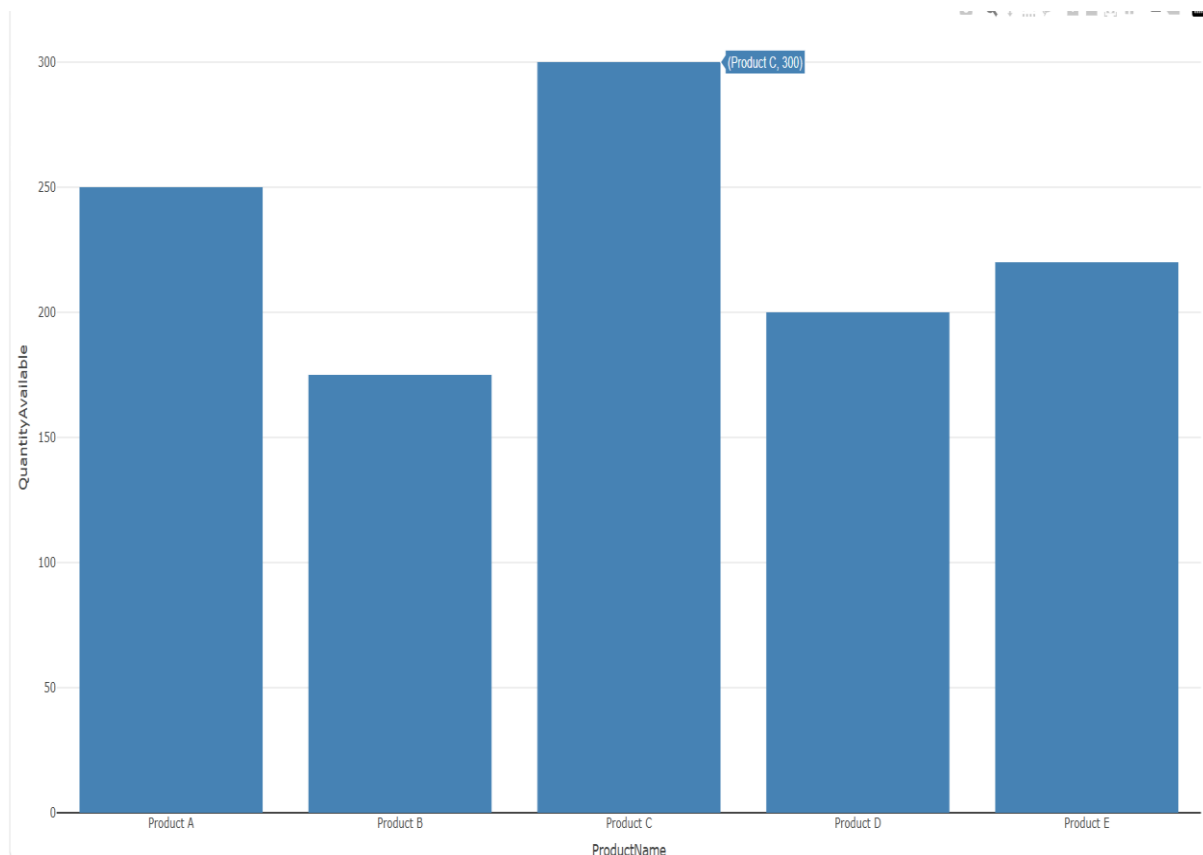
```

type="bar",
marker=list(color="steelblue"))

fig

```

Output:



Set 5

Program 1: Line Chart – Daily Page Views Trend

```
Date <- c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")
```

```
PageViews <- c(1500,1600,1400,1650,1800)
```

```
CTR <- c(2.3,2.7,2.0,2.4,2.6)
```

```
data <- data.frame(Date, PageViews, CTR)
```

```
plot(PageViews,
```

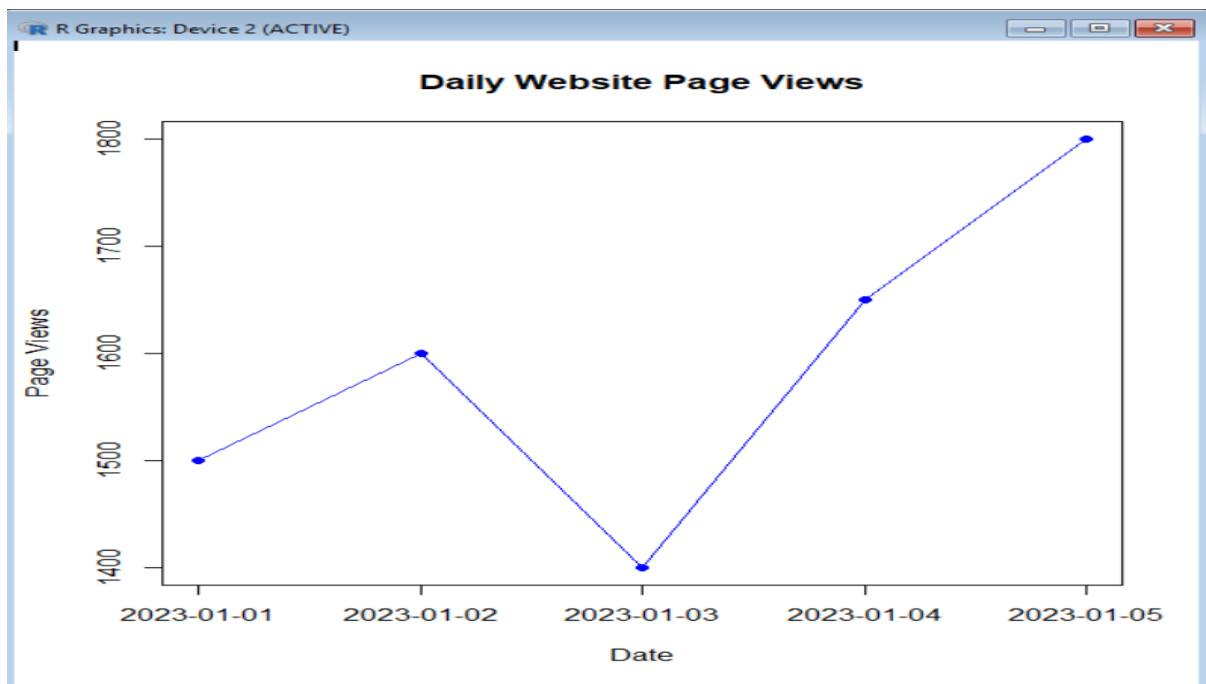
```
type="o",
```

```

col="blue",
pch=16,
xaxt="n",
xlab="Date",
ylab="Page Views",
main="Daily Website Page Views")
axis(1,
     at=1:5,
     labels=data$Date)

```

Output:



Program 2: Bar Chart – Click-through Rate (CTR)

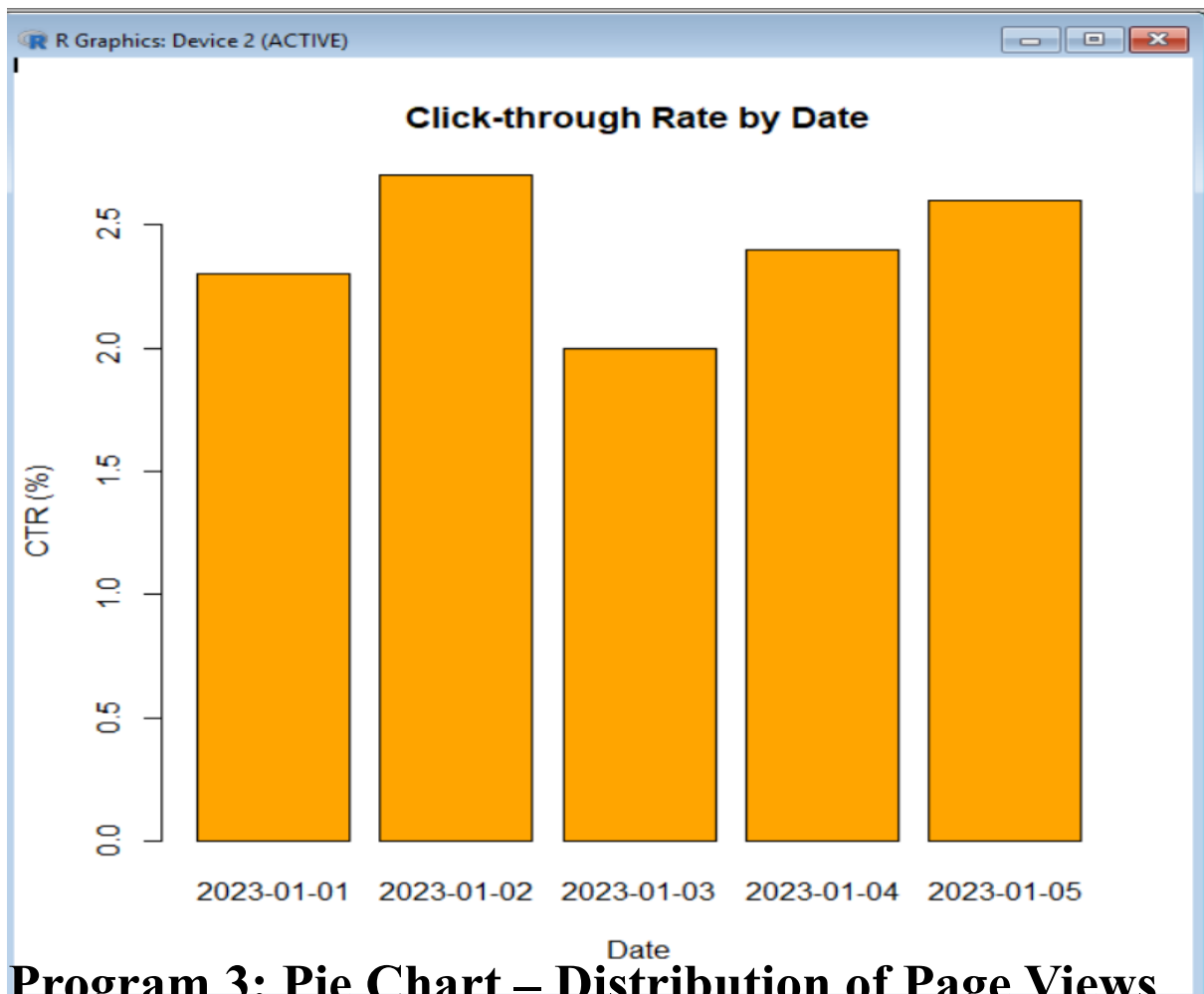
```

Date <- c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")
PageViews <- c(1500,1600,1400,1650,1800)
CTR <- c(2.3,2.7,2.0,2.4,2.6)
data <- data.frame(Date, PageViews, CTR)
barplot(data$CTR,
        names.arg=data$Date,
        col="orange",

```

```
main="Click-through Rate by Date",  
xlab="Date",  
ylab="CTR (%)")
```

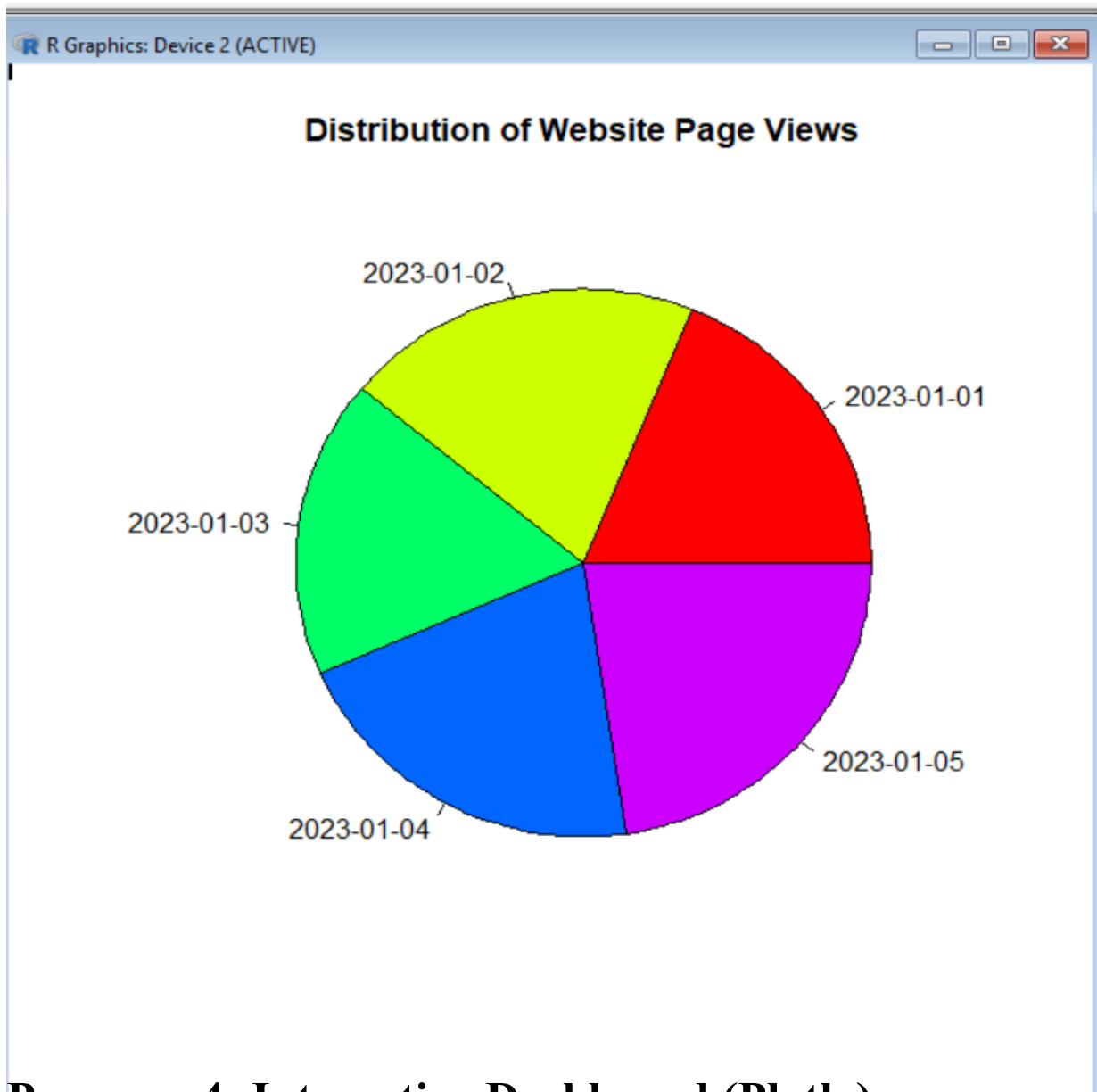
Output:



Program 3: Pie Chart – Distribution of Page Views

```
Date <- c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")  
PageViews <- c(1500,1600,1400,1650,1800)  
pie(PageViews,  
    labels=Date,  
    col=rainbow(5),  
    main="Distribution of Website Page Views")
```

Output:



Program 4: Interactive Dashboard (Plotly)

```
library(plotly)
Date <- c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")
PageViews <- c(1500,1600,1400,1650,1800)
CTR <- c(2.3,2.7,2.0,2.4,2.6)
data <- data.frame(Date, PageViews, CTR)
fig <- plot_ly(data,
               x=~Date,
```

```
y=~PageViews,  
type="scatter",  
mode="lines+markers",  
name="Page Views")
```

fig

Output:

